

Implantation

Curso de Ingeniería de Software



UNIVERSIDAD DE
COSTA RICA

SA-CIE

Carrera de
Informática Empresarial
Sede del Atlántico

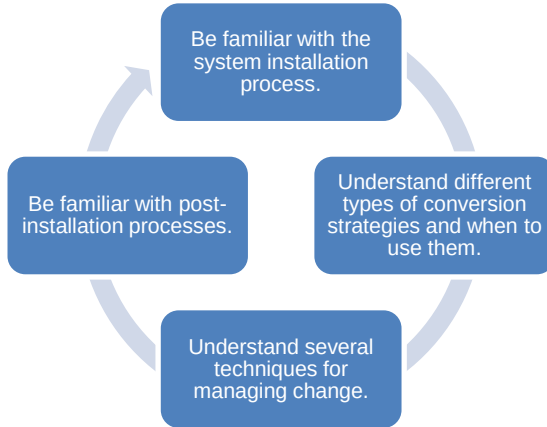
1. Introduction
2. Cultural Issues and Information Technology Adoption
3. Conversion
4. Change Management
5. Post-Implementation Activities

1. Introduction

Activities

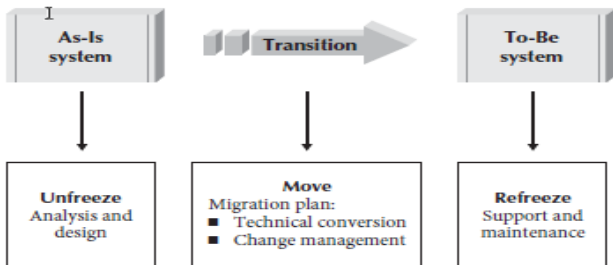
- The activities needed to install an information system and successfully convert an organization to using it.
- System support, system maintenance, and project assessment.
- Installing the system and making it available for use from a technical perspective are relatively straightforward.
- The training and organizational issues surrounding the installation are more complex and challenging because they focus on people, not computers.

Objectives



Change

- Managing the change to a new system is one of the most difficult tasks in any organization.
- As people use a computer system or set of work processes, those systems or work processes begin to become habits or norms; people learn them and become comfortable with them.



Change is a three-step process

1. Unfreeze: the existing habits and norms (the as-is system) so that change is possible
2. Move: to help the organization move to the new system via a migration plan. It has two major elements: One is technical, how the new system will be installed and how data in the as-is system will be moved into the to-be system. Another is organizational, helping users understand the change and motivating them to adopt it..
3. Refreeze: the new system as the habitual way of performing the work processes—ensuring that the new system successfully becomes the standard

- Change management is the most challenging of the three components because it focuses on people, not technology, is the least controllable by the project team.
- Change management means winning the hearts and minds of potential users and convincing them that the new system actually provides value
- Maintenance is the most costly aspect of the installation process.
- Project assessment enables project team members to step back and consider what they did right and what they could have done better

2. Cultural Issues and Information Technology Adoption

The cultural dimensions

Speed of messages: In a culture that values “deep” content, so that members of the culture can take their time to thoroughly understand the new system, simply providing an online help system is not going to be sufficient to ensure the successful adoption of the new information system.

Context: In high-context cultures, it is expected that the new information system will be placed into the entire context of the enterprise-wide system.

Time: In a polychronic time culture, the training could need to be spread out over a longer period of time, when compared to a monochronic time culture

Hofstede's dimensions

- Power distance, addresses how power issues are dealt with in the culture. For example, if a superior in an organization has an incorrect belief about an important issue, can a subordinate point out this error? In some cultures, the answer is a resounding no.
- Uncertainty avoidance, is based on the degree to which the culture depends on rules for direction, how well individuals in the culture handle stress, and the importance of employment stability.
- Individualism versus collectivism, is based on the level of emphasis the culture places on the individual or the collective. The relationship between the individual and the group is important for the success of an information system.

- Masculinity versus femininity, addresses how well masculine and feminine characteristics are valued by the culture. Some of the differences that could affect the adoption of an information system include employee motivational issues.
- Long- versus short-term orientation, if the local culture views success only in a short-term manner, then any new information system that is deployed to support one department of an organization may give that department a competitive advantage over other departments in the short run.

3. Conversion

- Conversion is the technical process by which a new system replaces an old system.
- The migration plan specifies what activities will be performed when and by whom and includes both technical aspects (such as installing hardware and software and converting data from the as-is system to the to-be system) and organizational aspects (such as training and motivating the users to embrace the new system).
- Conversion refers to the technical aspects of the migration plan.

Steps to the conversion plan before commencement of operations

To buy and install any needed hardware. In many cases, no new hardware is needed, but sometimes the project requires new hardware such as servers, client computers, printers, and networking equipment.



To install the software. This includes the to-be system under development and, sometimes, additional software that must be installed to make the system operational. At this point, the system is usually tested again to ensure that it operates as planned.



To convert the data from the as-is system to the to-be system. Data conversion is usually the most technically complicated step in the migration plan. Often, separate programs must be written to convert the data from the as-is system to the new formats required in the to-be system and store it in the to-be system files and databases. This process is often complicated by the fact that the files and databases in the to-be system do not exactly match the files and databases in the as-is system

3.1. Conversion Style



Direct Conversion: (sometimes called cold turkey, big bang, or abrupt cutover), the new system instantly replaces the old system. The new system is turned on, and the old system is immediately turned off. Is the simplest and most straightforward, but it's risky



Parallel Conversion: the new system is operated side by side with the old system; both systems are used simultaneously. After some time period (often one to two months) of parallel operation and intense comparison between the two systems, the old system is turned off and the organization continues using the new system. This approach is more likely to catch any major bugs in the new system and prevent the organization from suffering major problems.

3.2. Conversion Location

Pilot Conversion:

- one or more locations or units or work groups within a location are selected to be converted first as part of a pilot test. If the system passes the pilot test, then the system is installed at the remaining locations. Pilot conversion has the advantage of providing an additional level of testing before the system is widely deployed throughout the organization. Requires more time before the system is installed at all organizational locations.

Phased Conversion:

- the system is installed sequentially at different locations. A first set of locations is converted, then a second set, then a third set, and so on, until all locations are converted.

Simultaneous Conversion:

- all locations are converted at the same time. The new system is installed and made ready at all locations; at a preset time, all users begin using the new system. Simultaneous conversion eliminates problems with having different organizational units using different systems and processes. Must have sufficient staff to perform the conversion and train the users at all locations simultaneously.

3.3 Conversion Modules



Whole-System Conversion: the entire system is installed at one time, is the most common.



Modular Conversion: When the modules within a system are separate and distinct, organizations sometimes choose to convert to the new system one module at a time—i.e., using modular conversion. Each module either must be written to work with both the old and new systems or object wrappers. When there is only a loose association between modules, module conversion is easier



3.4 Selecting the Appropriate Conversion Strategy - Risk

- After the system has passed a rigorous battery of unit, system, integration, and acceptance testing, it should be bug free . . . maybe. Because humans make mistakes, nothing built by people is ever perfect.
- The conversion process provides one last step in which to catch these bugs before the system goes live and the bugs have the chance to cause problems.
- Parallel conversion is less risky than is direct conversion because it has a greater chance of detecting bugs that have gone undiscovered in testing.
- Risk is likely to be a very important factor in the conversion process if the system has not been as thoroughly tested as it might have been or if the cost of bugs is high.

3.4 Selecting the Appropriate Conversion Strategy - Cost

- Different conversion strategies have different costs. These costs can include things such as salaries for people who work with the system, travel expenses, operation expenses, communication costs, and hardware leases.
- Parallel conversion is more expensive than direct cutover because it requires that two systems be operated at the same time.
- Pilot conversion and phased conversion have somewhat similar costs. Simultaneous conversion has higher costs because more staff are required to support all the locations as they simultaneously switch from the old to the new system. Modular conversion is more expensive than whole-system conversion because it requires more programming.

3.4 Selecting the Appropriate Conversion Strategy - Time required

- The final factor is the amount of time required to convert between the old and the new system.
- Direct conversion is the fastest because it is immediate.
- Parallel conversion takes longer because the full advantages of the new system do not become available until the old system is turned off.
- Simultaneous conversion is fastest because all locations are converted at the same time. Phased conversion usually takes longer than pilot conversion because once the pilot test is complete all remaining locations are usually (but not always) converted simultaneously.
- Modular conversion takes longer than whole-system conversion because the models are introduced one after another.

Characteristic	Conversion Style		Conversion Location		
	Direct Conversion	Parallel Conversion	Pilot Conversion	Phased Conversion	Simultaneous Conversion
Risk	High	Low	Low	Medium	High
Cost	Low	High	Medium	Medium	High
Time	Short	Long	Medium	Long	Short

Conversion Modules	
Whole-System Conversion	Modular Conversion
High	Medium
Medium	High
Short	Long

4. Change Management

- Change management is the process of helping people to adopt and adapt to the to-be system and its accompanying work processes without undue stress.
- Resistance to change is common in most organizations. Therefore, the change management plan is an important part of the overall installation plan that glues together the key steps in the change management process. Successful change requires that people want to adopt the change and are able to adopt the change.

Key roles in any major organizational change

The first is the sponsor of the change—the person who wants the change. Usually, the sponsor is a senior manager of the part of the organization that must adopt and use the new system. It is critical that the sponsor be active in the change management process because a change that is clearly being driven by the sponsor



The second role is that of the change agent—the person(s) leading the change effort. The change agent, charged with actually planning and implementing the change, is usually someone outside of the business unit adopting the system and therefore has no direct management authority over the potential adopters.



The third role is that of potential adopters, or targets of the change—the people who actually must change. These are the people for whom the new system is designed and who will ultimately choose to use or not use the system.

4.1 Understanding Resistance to Change

- People resist change—even change for the better—for very rational reasons.
- What is good for the organization is not necessarily good for the people who work there.
- Learning to use the new system and work processes—even if the change is minor—requires more effort than continuing to use the existing, well-understood system and work processes.
- Every change has a set of costs and benefits associated with it. If the benefits of accepting the change outweigh the costs of the change,
- then people change.

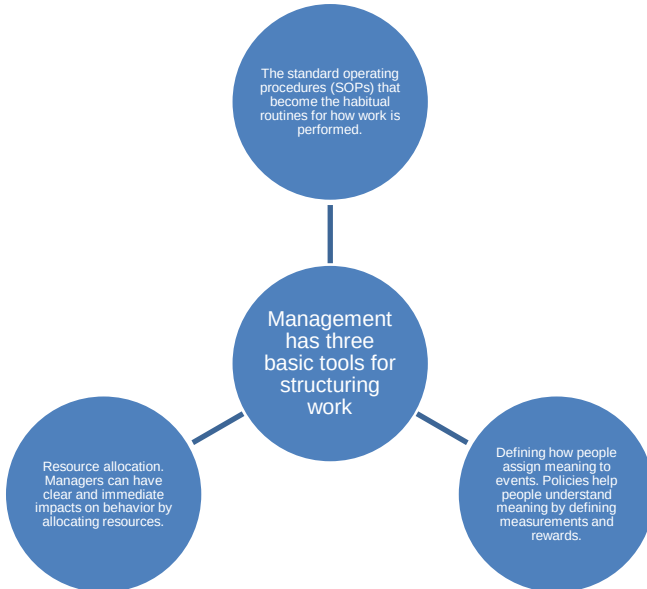
4.1 Understanding Resistance to Change (II)

- People perform a cost-benefit analysis and decide the extent to which they will embrace and adopt the change.
- They identify the costs of and benefits from the system and decide whether the change is worthwhile.
- There is some uncertainty as to whether a certain benefit or cost will actually occur
- There are also costs and, sometimes, benefits associated with the actual transition process itself. For example, suppose we found a nicer house, we might decide not to move simply because the cost of moving.

4.1 Understanding Resistance to Change (III)

- Adopting a new computer system might require us to learn new skills
- The first step in change management is to understand the factors that inhibit change—the factors that affect the perception of costs and benefits and certainty that they will be generated by the new system.
- People act on what they believe to be true, not on what is true. Thus, any understanding of how to motivate change must be developed from the viewpoint of the people expected to change, not from the viewpoint of those leading the change.

4.2 Revising Management Policies



4.3 Assessing Costs and Benefits

The first list is developed from the perspective of the organization, which should flow easily from the business case developed during the feasibility study and refined over the life of the project.

The second list of costs and benefits is developed from the viewpoints of the different potential adopters expected to change, or stakeholders in the change. Each of these potential adopters, or stakeholders, may have a different set of costs and benefits associated with the change.

The next step in developing a change management plan is to develop two clear and concise lists of costs and benefits provided by the new system compared with the as-is system.

4.3 Assessing Costs and Benefits (II)

- An analysis of the costs and benefits for each set of potential adopters, or stakeholders, will help pinpoint those who will likely support the change and those who might resist the change.
- The challenge at this point is to try to change the balance of the costs and benefits for those expected to resist the change so that they support it.
- All change is made by individuals, not organizations. There must also be a compelling reason for the organization to need the change.
- A clear migration plan developed by a credible change agent who has support from the business sponsor

4.4. Motivating Adoption

- Everyone who is expected to adopt the change must be convinced that the benefits from the to-be system outweigh the costs of changing.
- The goal of change management is to actively support and encourage the ready adopters and help them win over the reluctant adopters. it is often best to ignore this small minority of resistant adopters and focus on the larger majority of ready and reluctant adopters.

4.4. Motivating Adoption - basic strategies



Informational: the goal is to convince potential adopters that the change is for the better. Using this approach, the project team provides clear and convincing evidence of the costs and benefits of moving to the to-be system. This information is disseminated widely throughout the target group, much like an advertising or public relations campaign



Political: With a political strategy, organizational power, not information, is used to motivate change. This approach is often used when the cost-benefit set of the target adopters has more costs than benefits. The political strategy is usually beyond the control of the project team. It requires someone in the organization who holds legitimate power over the target group to influence the group to adopt the change.

4.5 Enabling Adoption: Training

- Potential adopters might want to adopt the change, but unless they are capable of adopting it, they won't.
- Careful training enables adoption by providing the skills needed to adopt the change.
- We have found that training is one of the most commonly overlooked parts of the process.
- Every new system requires new skills, either because the basic work processes have changed.
- In general, there are three ways to get these new skills:
 - To hire new employees who have the needed skills that the existing staff does not.
 - To outsource the processes to an organization that has the skills that the existing staff does not.
 - Training existing staff in the new business processes and the to-be system.

What to Train

Training for business systems should focus on helping the users to accomplish their jobs, not on how to use the system. The system is simply a means to an end, not the end in itself. The training must help the users understand how the computer fits into the bigger picture of their jobs.

The training should focus on what the user needs to do, not what the system can do. Most systems provide far more capabilities than the users will need to use. Rather than attempting to teach the users all the features of the system, training should instead focus on the much smaller set of activities that users perform on a regular basis and ensure that users are truly expert in those.

How to Train

There are many ways to deliver training. The most commonly used approach is classroom training, in which many users are trained at the same time by the same instructor.

It is also possible to provide one-on-one training, in which one trainer works closely with one user at a time. This is obviously more expensive

Another approach that is becoming more common is to use some form of computer-based training (CBT), in which the training program is delivered via computer

If there are only a few users to train, one-on-one training is the most effective. If there are many users to train, many organizations turn to CBT.

Quite often, large organizations use a combination of all three methods. Regardless of which approach is used, it is important to leave the users with a set of easily accessible materials that can be referred to long after the training has ended

5. Post-Implementation Activities

Institutionalization of the use of the new system—i.e., to make it the normal, accepted, routine way of performing the business processes.

5.1. System Support

Once the project team has installed the system and performed the change management activities, the system is officially turned over to the operations group.

Members of the operations group are usually closely involved in the installation activities because they are the ones who must ensure that the system actually works.

Providing system support means helping the users to use the system. Usually, this means providing answers to questions and helping users understand how to perform a certain function.

Online support is the most common form of on-demand training. This includes the documentation and help screens built into the system.

The help desk supports all systems, not just one specific system, so it receives calls about a wide variety of software and hardware.

5.1. System Support (II)

The level-2 support staff members are people who know the application system well and can provide expert advice. Sometimes the level-2 support staff members participate in training during the change management process to become more knowledgeable about the system, the new business processes, and the users themselves.

The level-2 support staff works with users to resolve problems. However, sometimes, particularly in the first few months after the system is installed, the problem turns out to be a bug in the software that must be fixed.

5.2. System Maintenance

Is the process of refining the system to make sure it continues to meet business needs.

More money and effort are devoted to system maintenance than to the initial development of the system, simply because a system continues to change and evolve as it is used.

Most beginning systems analysts and programmers work first on maintenance projects.

Whenever a potential change to the system is identified, a change request is prepared and forwarded to the project manager. It describes the change requested and explains why the change is important.

There is an initial assessment of feasibility and of costs and benefits, and the change request is prioritized. Then a systems analyst performs the analysis, which might include interviewing users, and prepares an initial design before programming begins.

The new (or revised) program is then extensively tested before the system is converted from the old system to the revised one.

5.2. System Maintenance - sources

Problem reports from the operations group that identify bugs in the system that must be fixed. These are usually given immediate priority because a bug can cause significant problems.

Enhancement to the system from users. Such enhancements are important in satisfying the users and are often key in ensuring that the system changes as the business requirements change.

Need to integrate two systems are generally rare but are becoming more common as system integration efforts become more common.

Those that occur when underlying software or networks change. Although users might never see these changes (because most changes are inside the system and do not affect its user interface or functionality).

Senior management, often driven by major changes in the organization's strategy or operations. These significant change requests are typically treated as separate projects, but the project manager responsible for the initial system is often placed in charge of the new project.

Problem Reports

Search: View:

Test ID: Requested information:

Objective:

Expected results:

Test cases:

Test ID	Test Case	Test Result
1. 10000001	Test case 1	Pass
2. 10000002	Test case 2	Fail
3. 10000003	Test case 3	Pass
4. 10000004	Test case 4	Fail
5. 10000005	Test case 5	Pass
6. 10000006	Test case 6	Fail

Report

Expected results:

Test cases:

Actual results:

Test cases:



Users

Change Request

Search: View:

Test ID: Requested information:

Objective:

Expected results:

Test cases:

Test ID	Test Case	Test Result
1. 10000001	Test case 1	Pass
2. 10000002	Test case 2	Fail
3. 10000003	Test case 3	Pass
4. 10000004	Test case 4	Fail
5. 10000005	Test case 5	Pass
6. 10000006	Test case 6	Fail

Report

Expected results:

Test cases:

Actual results:

Test cases:



Software or Network Changes

Problem Reports

Changes to Other Systems

1. Potential Change

2. Change Request with Feasibility, Costs, and Benefits

3. Priority

Project Manager

4. Change Request

6. Changed System

5. Design



Programmer



Analyst

Change Committee



5.3. Project Assessment

- The goal of project assessment is to understand what was successful about the system and the project activities and what needs to be improved.
- Nonetheless, assessment can be an important component in organizational learning because it helps organizations and people understand how to improve their work.

Parts to project assessment

Project Team Review:

- Focuses on the way the project team carried out its activities.
- The focus is on performance improvement, not penalties for mistakes made.
- By explicitly identifying mistakes and understanding their causes, project team members will, it is hoped, be better prepared for the next time they encounter a similar situation.
- The project manager then prepares a summary document that outlines the lessons learned from the project.

System Review

- The focus is to understand the extent to which the proposed costs and benefits from the new system identified during feasibility analysis were actually recognized from the implemented system.
- system review is often undertaken several months after the system is installed because it often takes a while before the system can be properly assessed.
- System review starts with the system request and feasibility analysis prepared at the start of the project.
- This helps the organization assess whether the system actually provided the value it was planned to provide.

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